hCoCena: Horizontal integration and analysis of transcriptomics datasets

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Introduction

Transcriptome-based gene co-expression analysis has become a standard procedure for structured and contextualized understanding and comparison of different conditions and phenotypes. Since large study designs with a broad variety of conditions are costly and laborious, extensive comparisons are hindered when utilizing only a single data set. Thus, there is an increased need for tools that allow the integration of multiple transcriptomic data sets with subsequent joint analysis, which can provide a more systematic understanding of gene co-expression and co-functionality within and across conditions. To make such an integrative analysis accessible to a wide spectrum of users with differing levels of programming expertise it is essential to provide user-friendliness and customizability as well as thorough documentation.

Here, we introduce horizontal CoCena (hCoCena: horizontal construction of co-expression networks and analysis), an R-package for network-based co-expression analysis that allows the analysis of a single transcriptomic data set as well as the joint analysis of multiple data sets. With hCoCena we provide a freely available, user-friendly, and adaptable tool for integrative multi-study or single-study transcriptomics analyses.

The hCoCena R-package is provided together with R Markdowns that implement an exemplary analysis workflow including extensive documentation and detailed descriptions of data structures and objects. A showcase example is also hosted on Fastgenomics to provide an interactive introduction to new users. Such efforts not only make the tool easy to use but also enable the seamless integration of user-written scripts and functions into the workflow, creating a tool that provides a clear design while remaining flexible and highly customizable. The package and additional information including an extensive Wiki are freely available on GitHub: https://github.com/MarieOestreich/hCoCena.

1 Main Phases of the Workflow

A three-step concept has been chosen for the presented tool: The pre-integration phase, which contains pre-processing steps that are dataset-specific, followed by the network-based integration phase and the post-integration phase, which conducts different cross-layer analyses

2 GFC calculation

Group Fold-Changes (GFCs) reflect the overall expression trend of a gene with respect to groups of samples that form disjoint subsets of the entirety of samples (e.g., a treatment, a disease).

The calculation of the GFCs is different depending on the absence or presence of control samples in the data sets.

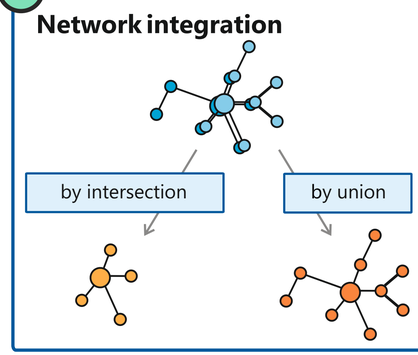
Without controls:

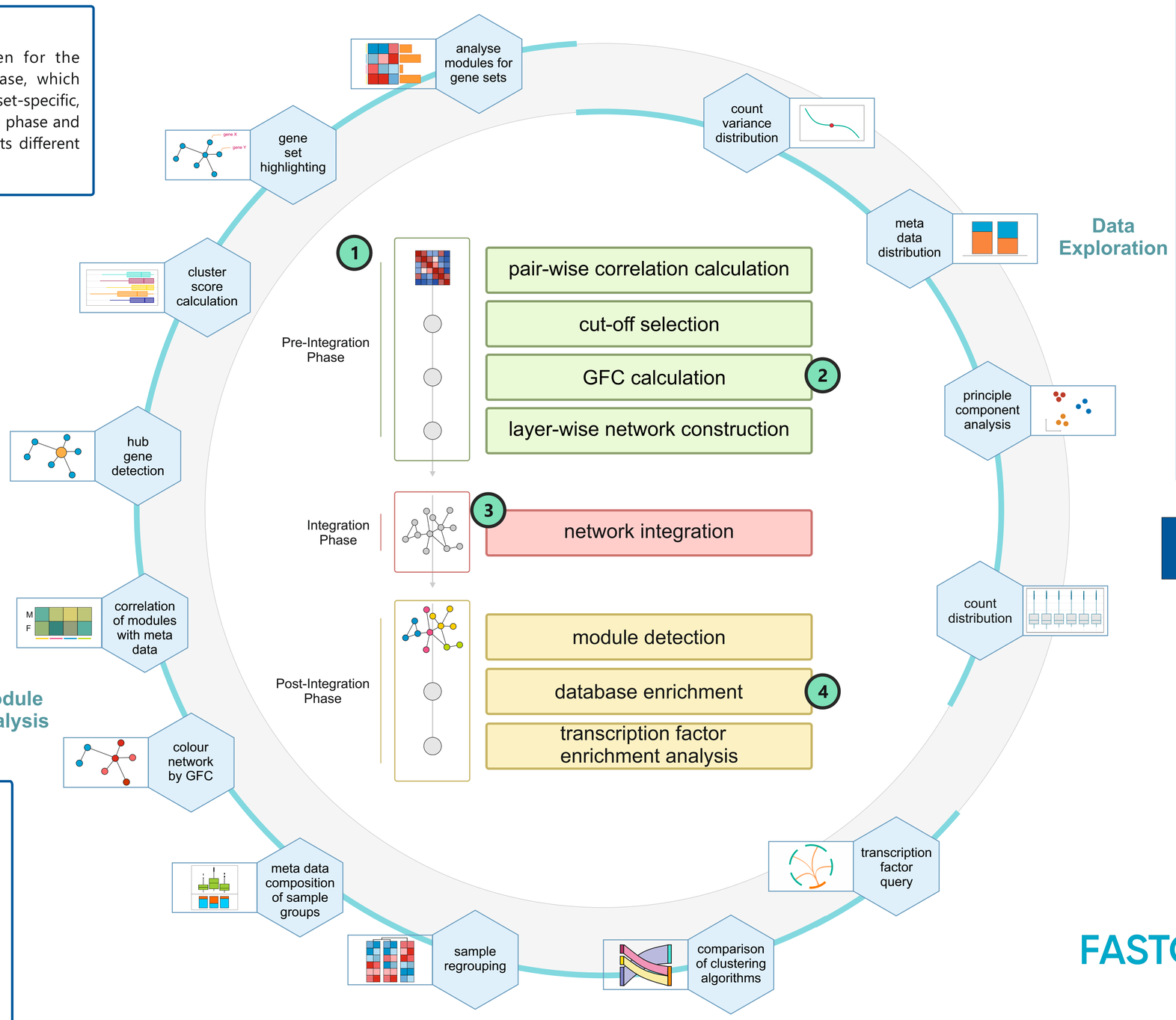
$$\frac{mean\left(g_{x_i}\right)}{\frac{1}{|X|} * \sum_{x_i \in X} mean\left(g_{x_i}\right)}$$

With controls:

$$\frac{mean\left(g_{x_{i,n}}\right)}{mean\left(g_{x_{c,n}}\right)}$$

3 Network integration





4 Database Enrichment

GENEONTOLOGY Unifying Biology

KEGG

MSigDB Molecular Signatures Database

reactome

Availability

GitHub

zenodo

FASTGENOMICS

Conclusion

The wealth of transcriptomic datasets that has been accumulated over the past years and continues to increase has demanded the development of tools that allow the integrative analysis of this data. With hCoCena, we offer a solution to this problem, by combining different transcriptomic datasets using a network-based integration approach. The tool further provides the user with a wide variety of downstream analysis applications and makes it easy to add additional functionalities specifically designed for the data at hand.

hCoCena is the first step to what is intended to become a tool suite for omics data analysis offering options for single- and multi-set analyses – as now provided by hCoCena – as well as planned extensions for single-cell data, multi-omics-data integration, and clinical study-oriented analysis. The suite is intended to assist researchers in easily tailoring an analysis workflow to their experimental setup, especially in the prospect of increased utilization of multi-omics technologies and the combined evaluation of the retrieved data.

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